Calculus II Assignment 8

Part I. Regular Problems

1. Use the Chain Rule to find dz/dt or dw/dt.

$$z = xy^3 - x^2y, \quad x = t^2 + 1, \quad y = t^2 - 1$$

2. Use the Chain Rule to find $\partial z/\partial s$ and $\partial z/\partial t$.

$$z = \tan^{-1}\left(x^2 + y^2\right), \quad x = s\ln t, \quad y = te^x$$

3.
$$f(x, y, z) = y^2 e^{xyz}, \quad P(0, 1, -1), \quad \mathbf{u} = \left(\frac{3}{13}, \frac{4}{13}, \frac{12}{13}\right)$$

   (a) Find the gradient of f.

   (b) Evaluate the gradient at the point P.

   (c) Find the rate of change of f at P in the direction of the vector $\mathbf{u}$.

4. (a) Show that a differentiable function f decreases most rapidly at $\mathbf{x}$ in the direction opposite to the gradient vector, that is, in the direction of $-\nabla f(\mathbf{x})$.

   (b) Use the result of part (a) to find the direction in which the function $f(x, y) = x^4y - x^2y^3$ decreases fastest at the point $(2, -3)$.

5. Find an equation for the plane tangent to the graph of $f(x, y) = x + y^2$,

   (a) at $(x, y) = (0, 0)$,

   (b) at $(x, y) = (1, 2)$.

Part II. Group Problems

1. Find the following derivatives or partial derivatives.

   (a) $\frac{\mathrm{d}}{\mathrm{d}x}(f(x))^3$, where f is a differentiable function from $\mathbb{R}$ to $\mathbb{R}$

   (b) $\frac{\partial}{\partial x}\left(y + x^2\right)^3$

   (c) $\frac{\partial}{\partial x} g\left(y + x^2\right)$, where g is a differentiable function from $\mathbb{R}$ to $\mathbb{R}$

   (d) $\frac{\partial}{\partial y}(f(x, y))^3$, where f is a differentiable function from $\mathbb{R}$ to $\mathbb{R}$.

2. Find the derivatives f_{xx}, f_{xy}, and f_{yy}.

(a) $f(x, y) = x^2 - y^2$

(b) $f(x, y) = (x + y)^2$

(c) $f(x, y) = \mathrm{e}^{-x} \cos y$